

BANARAS HINDU UNIVERSITY

B.Voc. Semester I Examination, 2022-23

Paper: BVCA-102 — Fundamentals of Artificial Intelligence

Time: Three Hours

Maximum Marks: 70

Instructions: Answer All Sections. Candidates are required to write their answers in their own words as far as practicable.

SECTION — A

Note: Long Answer type questions to be answered in about 750 words each. Attempt any two:

15 × 2 = 30 Marks**Question 1.**

- What is Artificial Intelligence? Discuss five applications of Artificial Intelligence.
- What are the challenges faced by Artificial Intelligence?

Question 2.

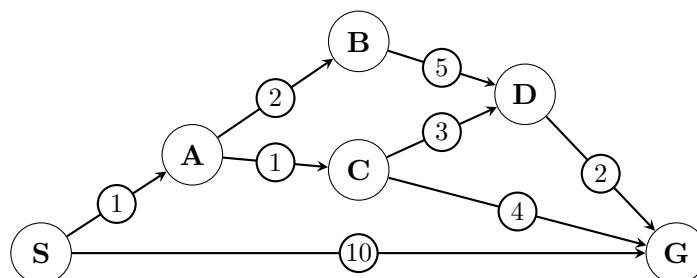
- What are agents and environment? What is an Ideal Rational Agent?
- Discuss different types of agents in Artificial Intelligence.

Question 3.

- What are Heuristic Search techniques? Discuss the Hill Climbing Algorithm.
- How does the Steepest Ascent Hill Climbing algorithm differ from the standard Hill Climbing algorithm?

Question 4.

- Explain AO* algorithm with example. [7]
- Discuss briefly A* algorithm. Find the optimal path and cost from Start node *S* to Goal node *G* for the given graph using A* algorithm. (The path cost is given on the edges of the graph and the table shows the values of heuristics cost). [8]



State	$h(n)$
S	5
A	3
B	4
C	2
D	6
G	0

SECTION — B

Note: Short Answer type questions to be answered in about 550 words each. (Any six)
($5 \times 6 = 30$)

Question 5. What are the steps for building an Expert System ?

Question 6. Discuss any five features of Environment.

Question 7. What do you understand by PEAS ? Design PEAS information for Self-driving cars and Medical Diagnostic System.

Question 8. What is Uninformed Search technique ? Discuss about Depth First Search algorithm.

Question 9. Define Facts, Rules and Queries in terms of Prolog Programming. Write a Prolog program to find maximum of two numbers.

Question 10. What is Means End Analysis ? Explain with example.

Question 11. Differentiate between Forward Chaining and Backward Chaining in the context of Expert Systems.

Question 12. What are Lists in prolog ? Define any three list operations.

SECTION — C

Question 13. Select the correct option: ($1 \times 10 = 10$)

(i) What is state space?

- (a) The whole problem
- (b) Your Definition to a problem
- (c) Problem you design
- (d) Representing your problem with variables and parameters

(ii) What is the space complexity of Depth-first search?

- (a) $O(b)$

- (b) $O(b^l)$
 - (c) $O(m)$
 - (d) $O(bm)$
- (iii) When is breadth-first search optimal?
- (a) When there is less number of nodes
 - (b) When all step costs are equal
 - (c) When all step costs are unequal
 - (d) None of the mentioned
- (iv) The father of Artificial Intelligence is:
- (a) Alan Turing
 - (b) John McCarthy
 - (c) Charles Babbage
 - (d) None of the above
- (v) A technique that was developed to determine whether a machine could or could not demonstrate artificial intelligence is known as:
- (a) Boolean Algebra
 - (b) Turing Test
 - (c) Logarithm
 - (d) Algorithm
- (vi) The component of an Expert system is:
- (a) Knowledge Base
 - (b) Inference Engine
 - (c) User Interface
 - (d) All of the above
- (vii) An AI agent perceives and acts upon the environment using:
- (a) Sensors
 - (b) Perceiver
 - (c) Actuators
 - (d) Both (a) and (c)
- (viii) The main function of a problem-solving agent is to:

- (a) Solve the given problem and reach the goal
 - (b) Find out which sequence of actions will get it to the goal state
 - (c) Both (a) & (b)
 - (d) None of the above
- (ix) Which of the following is not a variable in Prolog?
- (a) Paul
 - (b) a_123
 - (c) _123
 - (d) Hello
- (x) What would the output of the following Prolog query be?
- ?- f(a, b) = f(Y, X)
- (a) X=b, Y=a.
 - (b) This expression will not compile
 - (c) X=a, Y=b.
 - (d) X=X, Y=Y.